

# Megaloblastic — B12 Deficiency Causes





1. Vegan diet

WHAT YOU SEE

NUTRITIONAL row, vegan figure

WHAT IT ENCODES

B12 (cobalamin) is found only in animal products, so a strict vegan/vegetarian diet is the classic dietary cause. Because hepatic stores hold ~2–5 mg (years' supply) and daily needs are only ~2.4 µg, deficiency develops SLOWLY — typically over 3–5 years. Treat with oral or IM B12 supplementation; counsel pregnant/breastfeeding vegans because infants have minimal stores.

BOARD TRAP

Macrocytic anemia in a long-term vegan. WRONG answer: assume folate deficiency because the diet is plant-based. Discriminator: a plant-rich diet is usually folate-REplete; it is the lack of animal products that depletes B12 — check methylmalonic acid (elevated in B12, normal in folate deficiency) to separate them.

2. Pregnancy/infant brain

WHAT YOU SEE

Mother+baby '3-6 MO', 'BRAIN ATROPHY DELAYED MYELIN'

WHAT IT ENCODES

Infants of B12-deficient (often vegan or pernicious-anemia) mothers are born with low stores and become deficient around 3–6 months once maternal/breast-milk supply runs out. Consequences are neurologic: developmental regression, hypotonia, brain atrophy, and delayed myelination, which can be irreversible if untreated. Maternal repletion and infant supplementation are urgent.

BOARD TRAP

A 4-month-old breastfed infant of a vegan mother with regression and irritability. WRONG answer: attribute hypotonia to a primary neuromuscular disorder. Discriminator: timing at 3–6 months plus maternal dietary/PA history points to acquired infantile B12 deficiency — measure B12/MMA and treat promptly, because delay risks permanent neurodevelopmental damage.

3. Enterohepatic cycle

WHAT YOU SEE

ENTEROHEPATIC CYCLE arrows, 'SLOW DEPLETION'

WHAT IT ENCODES

B12 undergoes enterohepatic recirculation: bile secretes ~5–10 µg/day of B12 that is normally reabsorbed in the terminal ileum. This recycling, plus large hepatic stores, buffers intake gaps and explains the slow (years) onset of dietary deficiency. Crucially, in MALABSORPTION (e.g., ileal disease) both dietary AND recycled biliary B12 are lost, so deficiency develops much FASTER (months to ~1–2 years) than with poor diet alone.

BOARD TRAP

Comparing onset speed of dietary vs malabsorptive deficiency. WRONG answer: assume all B12 deficiency takes years. Discriminator: pure dietary lack is slow because the enterohepatic cycle is intact, but ileal malabsorption disrupts reabsorption of biliary B12 too — accelerating deficiency to months, especially after total gastrectomy or extensive ileal resection.

4. Pernicious anemia (gastric)

WHAT YOU SEE

GASTRIC row, stomach 'B12 TRAPPED', autoimmune

WHAT IT ENCODES

Pernicious anemia is autoimmune atrophic (type A) gastritis with destruction of parietal cells → loss of intrinsic factor (IF), so dietary B12 stays 'trapped' and unabsorbed. Antibodies: anti-intrinsic-factor (specific) and anti-parietal-cell (sensitive). It causes achlorhydria, hypergastrinemia, and a 2–3× increased risk of gastric adenocarcinoma and carcinoid; it associates with other autoimmune disease (thyroid, vitiligo, T1DM).

BOARD TRAP

Older patient with macrocytic anemia, paresthesias, and a positive anti-parietal-cell antibody only. WRONG answer: rely on parietal-cell antibody alone for diagnosis or skip cancer surveillance. Discriminator: anti-IF antibody is the SPECIFIC test (parietal-cell is sensitive but not specific); confirm with IF antibody, and remember PA mandates awareness of gastric cancer/carcinoid risk.

5. Gastrectomy / bariatric

WHAT YOU SEE

'TOTAL', 'PARTIAL 10-15%', bariatric surgery

WHAT IT ENCODES

Removing or bypassing parietal-cell-bearing stomach eliminates IF (and acid). Total gastrectomy causes inevitable B12 deficiency; partial gastrectomy and bariatric surgery (Roux-en-Y) cause it in ~10–15% and require lifelong B12 monitoring/supplementation. Onset is faster than dietary deficiency because both IF production and the acid needed to release food-bound B12 are lost.

BOARD TRAP

Anemia and neuropathy years after Roux-en-Y bypass. WRONG answer: blame iron deficiency alone. Discriminator: bariatric/gastric surgery causes multiple deficiencies, but loss of IF specifically impairs B12 — these patients need routine B12 (often IM or high-dose oral) prophylaxis, and B12/MMA should be checked alongside iron.

6. PPIs / atrophic gastritis

WHAT YOU SEE

PPI tablets, atrophic gastritis

WHAT IT ENCODES

Gastric acid and pepsin are needed to cleave B12 from dietary protein (food-bound B12). Chronic PPI/H2-blocker use and atrophic gastritis (autoimmune or H. pylori-related) reduce acid and impair this release, lowering B12 absorption over years. Notably, FREE/crystalline B12 in supplements does NOT require acid, so oral B12 tablets still work.

BOARD TRAP

Borderline-low B12 in a patient on long-term omeprazole. WRONG answer: stop the PPI as the definitive fix or assume oral B12 will fail. Discriminator: PPI-related deficiency affects FOOD-bound B12 because acid is needed for release — CRYSTALLINE oral B12 bypasses this and is absorbed, so supplementation works even if the PPI continues.



## 7. Terminal ileum disease/resection

**WHAT YOU SEE**

INTESTINAL row, ileocecal valve, Crohn's/resection

**WHAT IT ENCODES**

The IF–B12 complex is absorbed only in the TERMINAL ILEUM via cubam (cubilin/amnionless) receptors. Crohn's disease, ileal resection, radiation enteritis, or ileocecal disease therefore block absorption directly. Deficiency develops relatively quickly because both dietary and enterohepatically-recycled B12 are malabsorbed; these patients usually need PARENTERAL B12.

**BOARD TRAP**

Crohn's patient with prior ileal resection, now macrocytic anemia, given oral B12. WRONG answer: oral repletion in significant ileal disease. Discriminator: with terminal-ileum disease/resection the absorptive site is gone, so standard oral IF-dependent absorption fails — use IM B12 (or very-high-dose oral via passive diffusion). Ileal disease also impairs bile-salt reabsorption, hinting at the location.

## 8. Fish tapeworm

**WHAT YOU SEE**

'FISH TAPEWORM', raw fish '>1.2m', Diphylobothrium

**WHAT IT ENCODES**

Diphylobothrium latum, the fish tapeworm, is acquired from raw/undercooked freshwater fish (Scandinavia, Baltic, Great Lakes, Japan). The worm (which can exceed 1.2 m) avidly absorbs luminal B12, competing with the host and causing megaloblastic anemia. Diagnosis is by operculated eggs/proglottids in stool; treatment is praziquantel (or niclosamide) plus B12 repletion.

**BOARD TRAP**

Megaloblastic anemia in a patient who eats raw freshwater fish, with eosinophilia. WRONG answer: assume pernicious anemia by default. Discriminator: the dietary/exposure history (raw freshwater fish, endemic region) and stool ova/proglottids point to D. latum competing for B12 — antiparasitic therapy plus B12, not just IF-antibody workup.

## 9. Blind loop / bacterial overgrowth

**WHAT YOU SEE**

'STAGNANT GREEN POOL', diverticulitis, stricture, fistula, blind loop

**WHAT IT ENCODES**

Small-intestinal bacterial overgrowth (SIBO) — from blind loops, strictures, fistulae, diverticula, or dysmotility — lets bacteria CONSUME luminal B12 before absorption while paradoxically producing folate (so folate may be normal/high). It also causes bile-salt deconjugation → fat malabsorption. Diagnose with breath testing; treat the anatomic cause and give antibiotics (e.g., rifaximin) plus B12.

**BOARD TRAP**

Low B12 with NORMAL/high folate in a patient with scleroderma or surgical blind loop. WRONG answer: call it isolated dietary deficiency. Discriminator: the combination of low B12 with high folate is classic for SIBO, because bacteria consume B12 but synthesize folate — pursue breath testing/antibiotics, distinct from PA or dietary causes.

## 10. Celiac / tropical sprue

**WHAT YOU SEE**

Celiac, tropical plant, pancreas

**WHAT IT ENCODES**

Diffuse small-bowel mucosal disease impairs B12 handling: celiac disease and tropical sprue damage absorptive epithelium, and pancreatic exocrine insufficiency removes the proteases needed to release B12 from R-binder so it can bind IF. Pancreatic insufficiency gives a characteristic abnormal Schilling test that CORRECTS with pancreatic enzymes; tropical sprue responds to tetracycline + folate.

**BOARD TRAP**

Malabsorptive B12 deficiency that normalizes on the (historical) Schilling test only after adding pancreatic enzymes. WRONG answer: diagnose pernicious anemia. Discriminator: PA corrects with added IF; pancreatic insufficiency corrects with added ENZYMES (R-binder not cleaved); celiac/tropical sprue point to mucosal disease — the correction step localizes the lesion.

## 11. Metformin

**WHAT YOU SEE**

DRUGS row, 'METFORMIN' bottle, checkmark

**WHAT IT ENCODES**

Metformin is the most common drug cause of B12 deficiency. It interferes with the calcium-dependent uptake of the IF–B12 complex by ileal cubam receptors (effect partly reversible with oral calcium). Risk rises with dose and duration; guidelines suggest periodic B12 screening in long-term users, especially those with neuropathy.

**BOARD TRAP**

A diabetic on long-term metformin with peripheral neuropathy assumed to be purely diabetic. WRONG answer: attribute all neuropathy to diabetes. Discriminator: metformin-induced B12 deficiency can cause or worsen neuropathy — check B12/MMA in any long-term metformin user with neuro symptoms, since it is treatable and otherwise progressive.

## 12. Other drugs (PPI/PAS/colchicine)

**WHAT YOU SEE**

PAS, colchicine, neomycin, anticonvulsants, alcohol

**WHAT IT ENCODES**

Multiple drugs impair B12 absorption or metabolism: PAS (para-aminosalicylic acid), colchicine, and neomycin damage ileal absorption; chronic alcohol impairs absorption and storage; and anticonvulsants/oral contraceptives more often lower FOLATE. Nitrous oxide (separate box) inactivates B12 directly. A careful medication and alcohol history is essential in any megaloblastic anemia.

**BOARD TRAP**

Megaloblastic anemia in a patient on phenytoin assumed to be B12 deficiency. WRONG answer: replete B12 reflexively. Discriminator: anticonvulsants (phenytoin) classically cause FOLATE deficiency, whereas colchicine/neomycin/PAS impair B12 absorption — measure BOTH B12 and folate (and MMA/homocysteine) to identify which vitamin and which drug is responsible.



### 13. N2O abuse

#### WHAT YOU SEE

METABOLIC box B, 'N2O ABUSE', 'MMA NORMAL', neuropathy

#### WHAT IT ENCODES

Nitrous oxide irreversibly oxidizes the cobalt core of B12 ( $\text{Co}^{II} \rightarrow \text{Co}^{3+}$ ), inactivating methionine synthase and producing a FUNCTIONAL B12 deficiency even when serum B12 is normal. Recreational 'whippet' use or repeated anesthesia precipitates subacute combined degeneration: dorsal-column/lateral-column signs, paresthesias, and ataxia. Homocysteine and MMA rise; treat by stopping N2O and giving high-dose B12 ( $\pm$  methionine).

#### BOARD TRAP

Young patient with ataxia, paresthesias, and brisk reflexes; serum B12 is normal, so B12 deficiency is excluded. WRONG answer: rule out B12 because the serum level is normal. Discriminator: nitrous-oxide toxicity causes FUNCTIONAL deficiency with normal serum B12 — elevated MMA/homocysteine and the exposure history reveal it; treat as subacute combined degeneration despite the normal level.

### 14. TC II deficiency (infant)

#### WHAT YOU SEE

METABOLIC box A, baby 'weeks old', 'B12 NORMAL', irreversible neuro

#### WHAT IT ENCODES

Transcobalamin II (TC II) is the transport protein that delivers absorbed B12 to tissues. Hereditary TC II deficiency presents in the first WEEKS of life with severe megaloblastic anemia, failure to thrive, and neurologic injury despite NORMAL serum B12 (because the assay measures mostly haptocorrin-bound, inactive B12). It requires urgent high-dose parenteral B12; delay causes irreversible neurologic damage.

#### BOARD TRAP

Neonate with severe megaloblastic anemia but normal serum B12, deemed not B12-related. WRONG answer: exclude a cobalamin disorder based on normal serum B12. Discriminator: TC II deficiency causes a tissue-level (functional) deficiency with NORMAL serum B12 — the early-infancy onset and high MMA/homocysteine identify it; aggressive B12 prevents permanent neuro damage.

### 15. Cobalamin metabolism mutase defect

#### WHAT YOU SEE

METABOLIC box C, baby, 'ACIDOSIS', 'KETOSIS', 'NORMOCYTIC', 'MUTASE DEFECT', 'B12 OK'

#### WHAT IT ENCODES

Inherited defects of intracellular cobalamin processing or methylmalonyl-CoA mutase cause methylmalonic acidemia (and homocystinuria in some cbl subtypes). Infants present with metabolic ACIDOSIS, KETOSIS, vomiting, lethargy, and failure to thrive — but the anemia is typically NORMOCYTIC rather than megaloblastic, and serum B12 is normal. Some cobalamin-handling subtypes respond to pharmacologic hydroxocobalamin; isolated mutase apoenzyme defects do not.

#### BOARD TRAP

Acidotic, ketotic neonate with high MMA assumed to have dietary/absorptive B12 deficiency. WRONG answer: treat as ordinary B12 deficiency with a megaloblastic picture. Discriminator: a mutase/cobalamin-metabolism defect gives NORMAL serum B12, NORMOCYTIC blood, and a metabolic-acidosis/organic-aciduria picture — distinguish from acquired B12 deficiency, and trial B12-responsive subtypes with hydroxocobalamin.

### 16. Post-transplant GVHD / immune

#### WHAT YOU SEE

INTESTINAL row right end, 'IMMUNE', 'GVHD' with immune cells

#### WHAT IT ENCODES

Immune-mediated gut injury can cause B12 malabsorption: intestinal graft-versus-host disease after allogeneic HSCT and autoimmune enteropathy damage the absorptive mucosa (including the terminal ileum), impairing IF-B12 uptake. This adds to other post-transplant deficiencies and warrants B12 monitoring in patients with gut GVHD or chronic diarrhea after transplant.

#### BOARD TRAP

Allo-transplant patient with chronic diarrhea and macrocytic anemia attributed solely to medication marrow suppression. WRONG answer: blame only drug myelosuppression. Discriminator: intestinal GVHD damages the absorptive mucosa and can cause true B12 (and other vitamin) malabsorption — check B12/MMA, because this is a correctable contributor distinct from drug effect.